

## 2 Weighted Voting

### 2.3 Shapley-Shubik Power Index

**Definition 2.3.1** A sequential coalition is an ordered coalition, that is, it keeps track of the order in which each player joins the coalition.

We will use the notation

$$\langle P_1, P_2, P_3 \rangle$$

to denote sequential coalitions.

**Definition 2.3.2** A **pivotal player** is the player in the coalition which changes the coalition from a losing coalition to a winning one.

**Definition 2.3.3** (Shapley-Shubik power index) To compute the shapley-shubik power index:

- List all sequential coalitions
- Determine all pivotal players
- Count up how many times each player was critical
- convert these into fractions by deviding each count by the total number of sequential coalitions

**Example 2.3.4** Calculate the Shapley-Shubik Power Index for the following weighted voting system

$$[8 : 6, 3, 2].$$

**Example 2.3.5** Compute the Shapley-Shubik Power Index for the weighted voting system:

$$[36 : 20, 17, 15]$$

$$P_1 = 1/2, P_2 = 1/2, P_3 = 0.$$

**Example 2.3.6** Compute the Shapley-Shubik Power Index for the weighted voting system:

$$[3 : 2, 0.5, 0.5, 0.5]$$

**Example 2.3.7** Consider the weighted voting system given by

$$[q : 7, 3, 1].$$

Which values of  $q$  result in a dictator? (list all values)

**Example 2.3.8** In the Shapley-Shubik method, is it possible for a dummy to be a pivotal player? Why or why not?

No, if they were, they would be a critical player in a winning coalition.